

12.26 Car rental 2

In the light of the solution to the problem stated in Section 12.25, the company wants to consider where it might be most worthwhile to expand repair capacity. The weekly fixed costs, given below, include interest payments on the necessary loans for expansion.

The options are as follows:

1. Expand repair capacity at Birmingham by 5 cars per day at a fixed cost per week of £18 000.
2. Further expand repair capacity at Birmingham by 5 cars per day at a fixed cost per week of £8000.
3. Expand repair capacity at Manchester by 5 cars per day at a fixed cost per week of £20 000.
4. Further expand repair capacity at Manchester by 5 cars per day at a fixed cost per week of £5000.
5. Create repair capacity at Plymouth of 5 cars per day at a fixed cost per week of £19 000.

If any of these options is chosen, it must be carried out in its entirety, that is, there can be no partial expansion. Also, a further expansion at a depot can be carried out only if the first expansion is also carried out, so for example option (2) at Birmingham cannot be chosen unless option (1) is also chosen. If option (2) is chosen, thereby also choosing option (1), these count as two options. Similar stipulations apply regarding the expansions at Manchester. At most three of the options can be carried out.

12.27 Lost baggage distribution

A small company with six vans has a contract with a number of airlines to pick up lost or delayed baggage, belonging to customers in the London area, from Heathrow airport at 6 p.m. each evening. The contract stipulates that each customer must have their baggage delivered by 8 p.m. The company requires a model, which they can solve quickly each evening, to advise them what is the minimum number of vans they need to use and to which customers each van should deliver and in what order. There is no practical capacity limitation on each van. All baggage that needs to be delivered in a two-hour period can be accommodated in a van. Having ascertained the minimum number of vans needed, a solution is then sought, which minimises the maximum time taken by any van.

On a particular evening, the places where deliveries need to be made and the times to travel between them (in minutes) are given in Table 12.24. No allowance

Table 12.24

Heathrow	20	25	35	65	90	85	80	86	25	35	20	44	35	82
Harrow	15	35	60	55	57	57	85	90	25	35	30	37	20	40
Ealing	30	50	70	55	55	55	50	65	10	25	15	24	20	90
Holborn	45	60	53	55	47	55	55	47	12	22	20	12	10	21
Sutton	46	45	15	45	75	25	45	25	25	11	19	15	25	25
Dartford	15	15	15	15	25	25	25	25	45	65	53	43	63	70
Bromley	17	17	17	17	25	25	17	25	41	25	33	27	45	30
Greenwich	25	25	25	25	25	25	25	25	40	34	32	20	30	10
Barking	65	65	65	65	65	65	65	65	65	70	72	61	45	13
Hammersmith	20	20	20	20	20	20	20	20	20	20	8	7	15	25
Kingston	5	5	5	5	5	5	5	5	5	5	5	12	45	65
Richmond	14	14	14	14	14	14	14	14	14	14	14	14	34	56
Battersea	30	30	30	30	30	30	30	30	30	30	30	30	30	40
Islington	27	27	27	27	27	27	27	27	27	27	27	27	27	27
Woolwich														

is made for drop off times. For convenience, Heathrow will be regarded as the first location.

Formulate optimisation models that will minimise the number of vans that need to be used, and within this minimum, minimise the time taken for the longest time delivery.

12.28 Protein folding

This problem is based on one in the paper by Forrester and Greenberg (2008). It is a simplification of a problem in molecular biology. We take a protein as consisting of a chain of amino acids. For the purpose of this problem, the amino acids come in two forms: *hydrophilic* (waterloving) and *hydrophobic* (water hating). An example of such a chain is given in Figure 12.8. with the hydrophobic acids marked in bold.

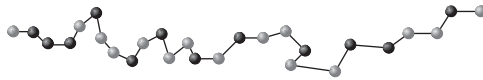


Figure 12.8

Such a chain naturally folds so as to bring as many hydrophobic acids, as possible, close together. An optimum folding for the chain, in two dimensions, is given in Figure 12.9, with the new matches marked by dashed lines. The problem is to predict the optimum folding. (Forrester and Greenberg also impose a condition that the resultant protein be confined to a given lattice of points. We do not impose that condition here). This problem can be modelled by a number of integer programming formulations. Some of these are discussed in the above reference. Another formulation is suggested in section 13.28. The problem posed here is to find the optimum folding for a chain of 50 amino acids with hydrophobic acids at positions 2, 4, 5, 6, 11, 12, 17, 20, 21, 25, 27, 28, 30, 31, 33, 37, 44 and 46 as shown in Figure 12.10.

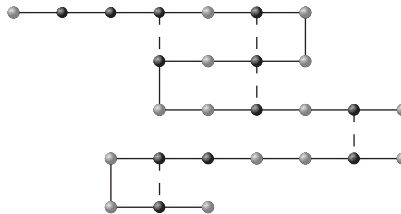


Figure 12.9